

1. Dataset Overview & Feature Engineering

Dataset shape : (109874, 35)

Working configs : 874 / 109,874

Datacenter exits: 592

Unique ASNs : 141

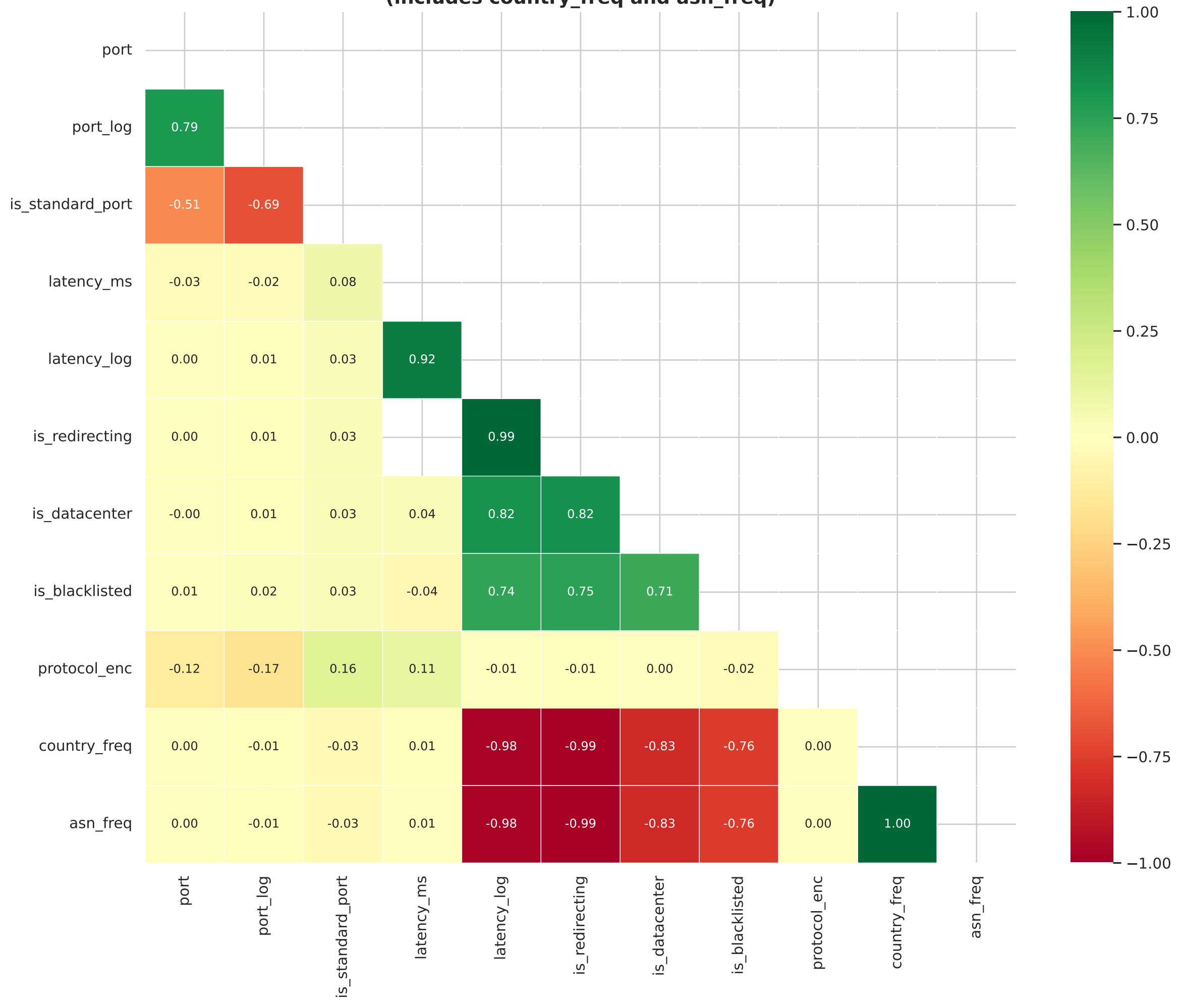
Security check coverage (working configs only):

proxy_detected : 445 / 874

has_ipv6 : 201 / 874

ipv6_leak : 32 / 874

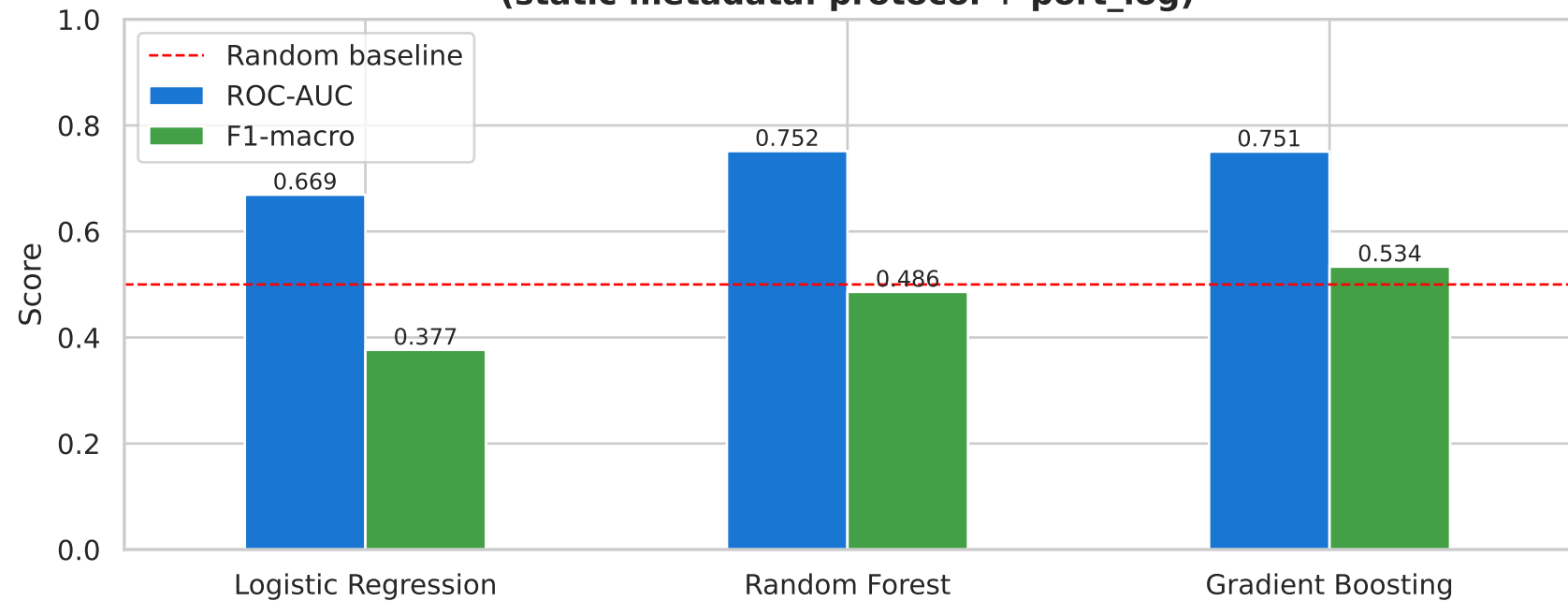
Feature Correlation Heatmap
(includes country_freq and asn_freq)



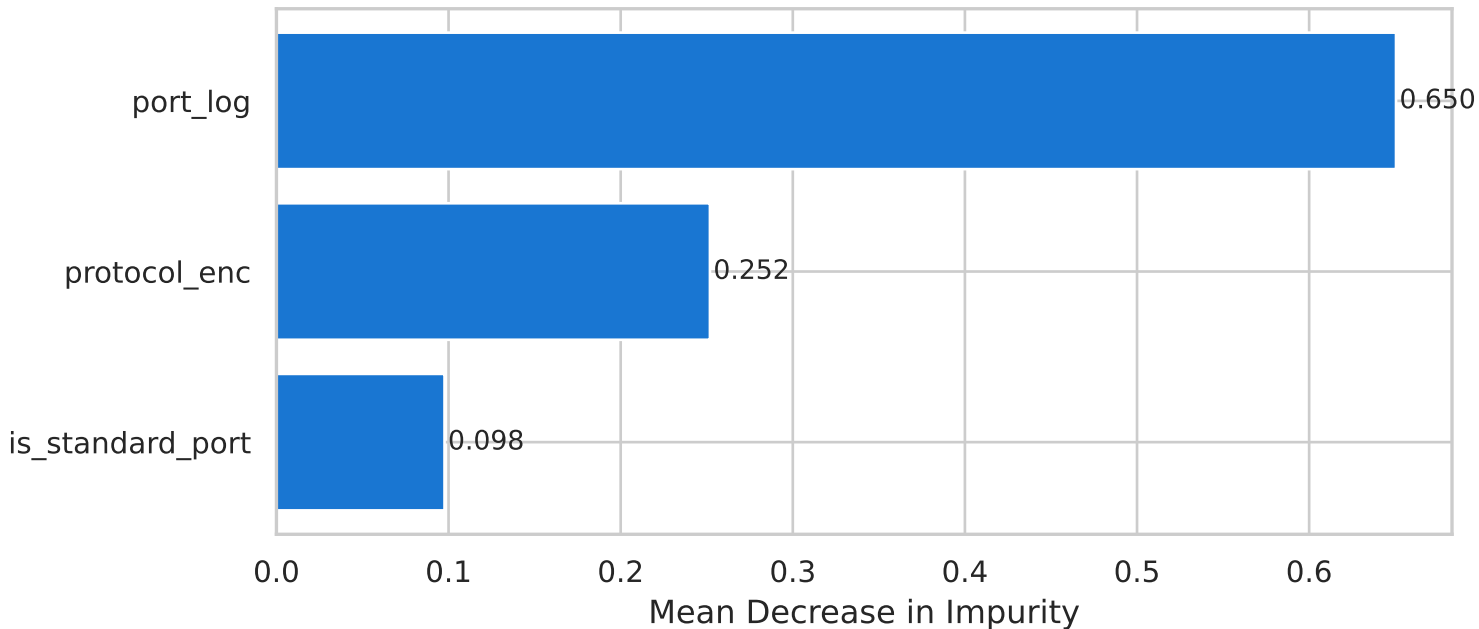
3. Availability Prediction — 5-Fold CV Scores

	ROC-AUC	F1-macro
Logistic Regression	0.669325	0.376659
Random Forest	0.751557	0.485983
Gradient Boosting	0.750605	0.533640

Availability Prediction — 5-Fold CV (static metadata: protocol + port_log)



Feature Importance — Predicting Availability (Random Forest, static metadata)

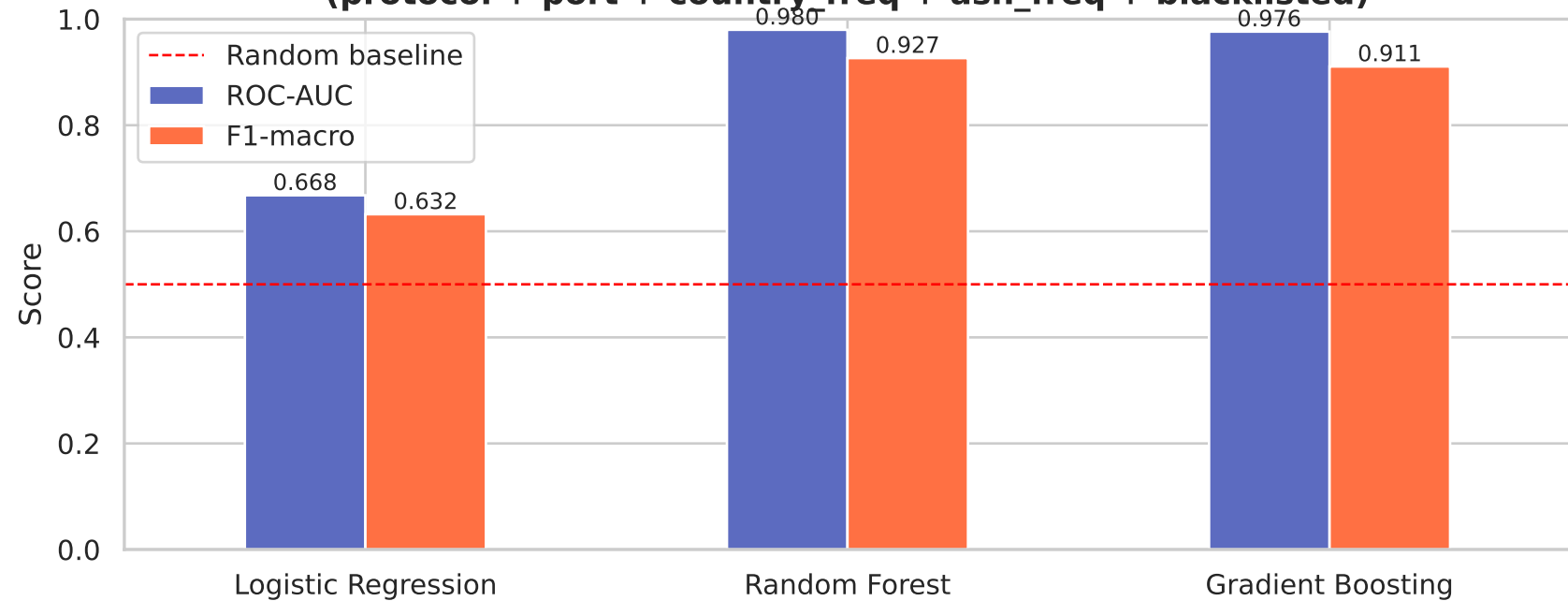


5. Datacenter Exit Prediction — CV Scores

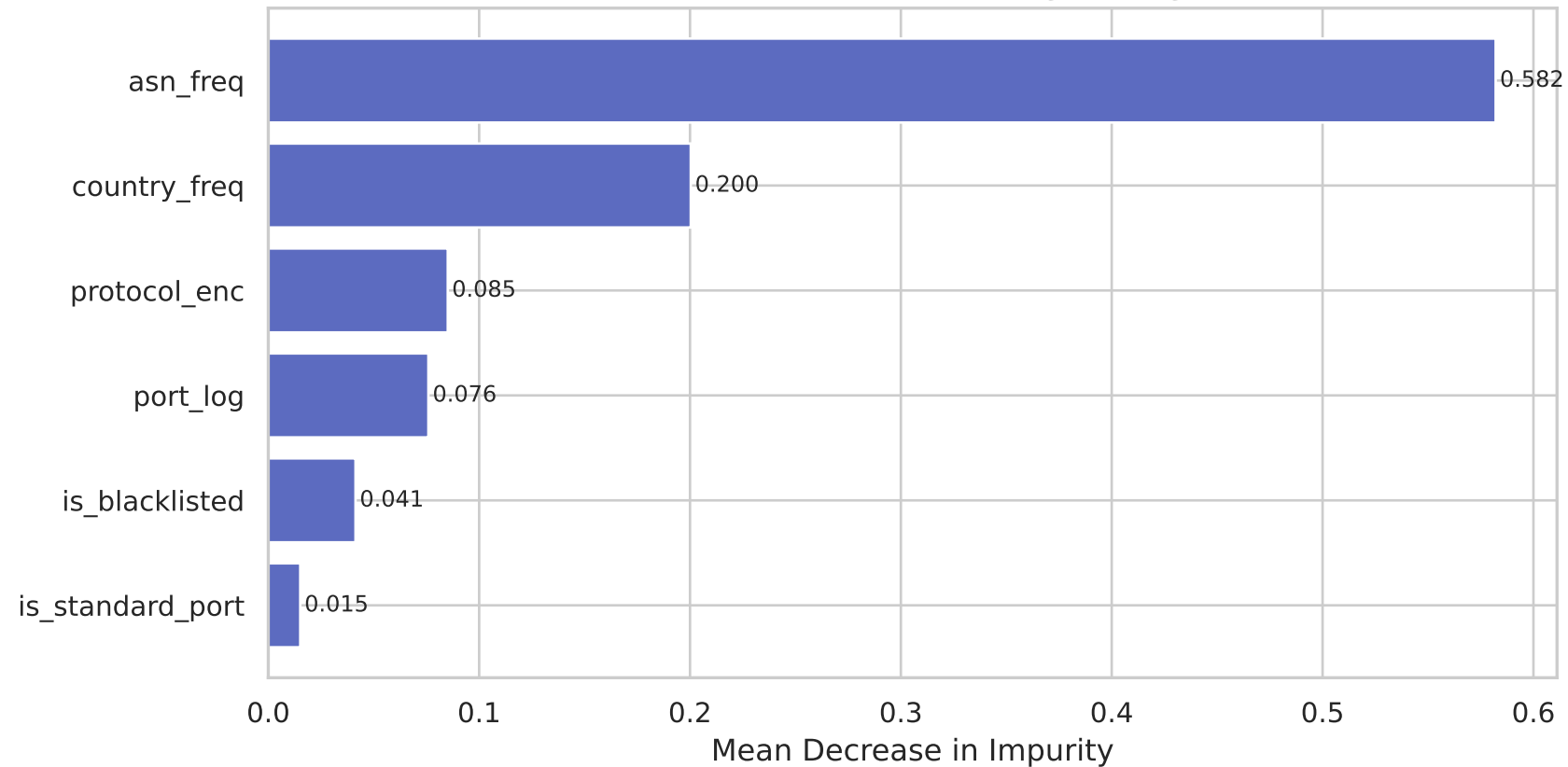
Dataset: 874 configs | datacenter=592 residential=282

	ROC-AUC	F1-macro
Logistic Regression	0.667857	0.632150
Random Forest	0.979914	0.926541
Gradient Boosting	0.976481	0.910710

Datacenter Exit Prediction — 5-Fold CV (protocol + port + country_freq + asn_freq + blacklisted)



Feature Importance — Predicting Datacenter Exit (Random Forest, working configs)



7. Latency Regression — Results

Latency Regression (log-scale, target-encoded country):

$$R^2 = -0.190 \pm 0.121$$

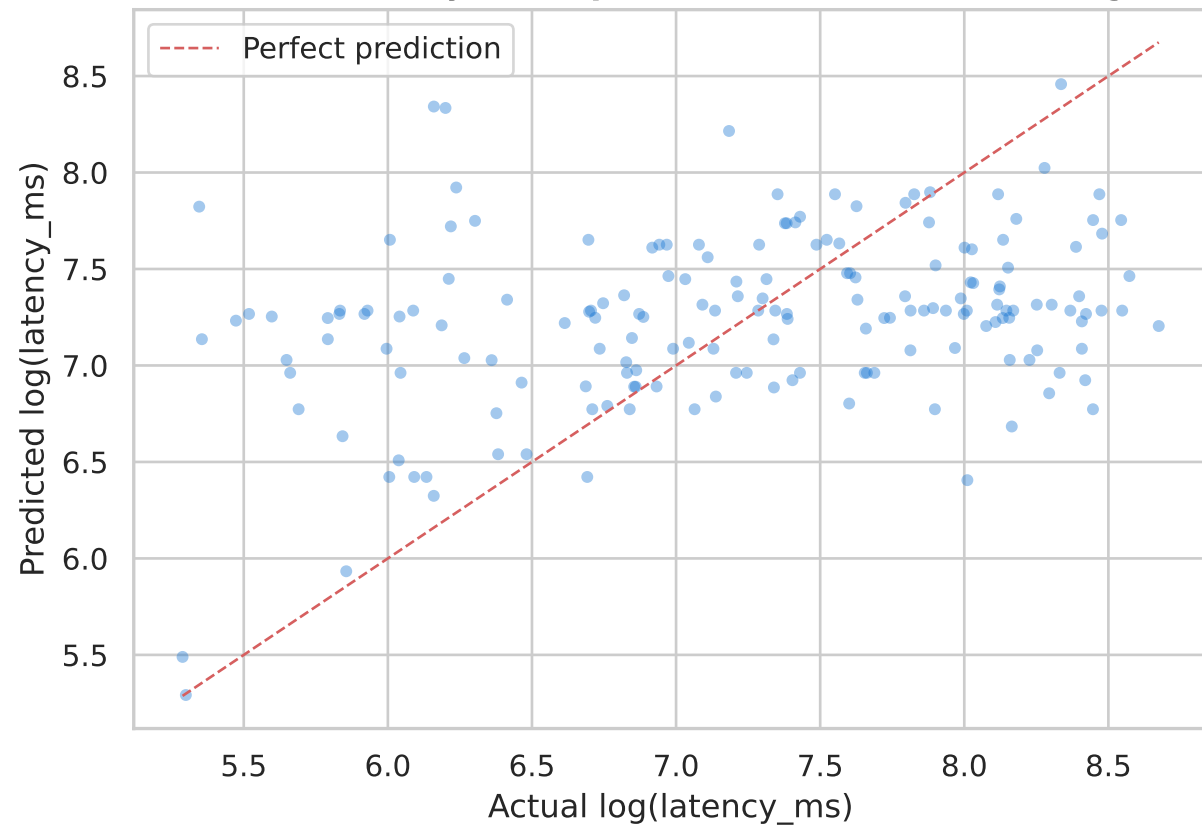
$$\text{MAE} = 0.805 \pm 0.076 \quad (\text{log-ms units})$$

$R^2 < 0$ means the model is worse than predicting the mean.

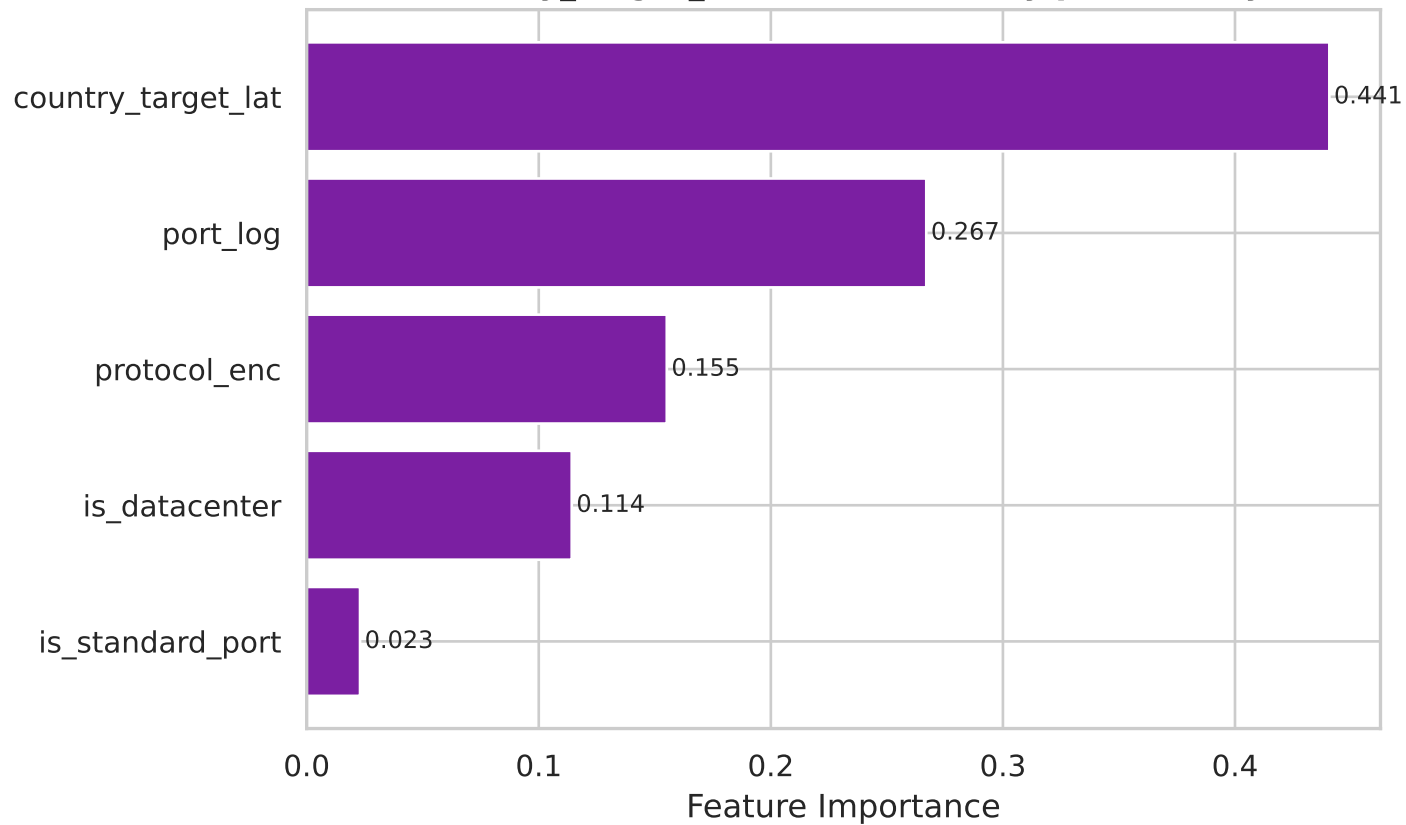
Latency is driven by real-time network conditions (server load, routing path, congestion) that are invisible in static config metadata. No amount of ML will fix this; the signal is not there.

Latency Regression — Test Set

CV $R^2=-0.190$ — latency is not predictable from static config metadata



Latency Feature Importance (country_target_lat = mean latency per country)



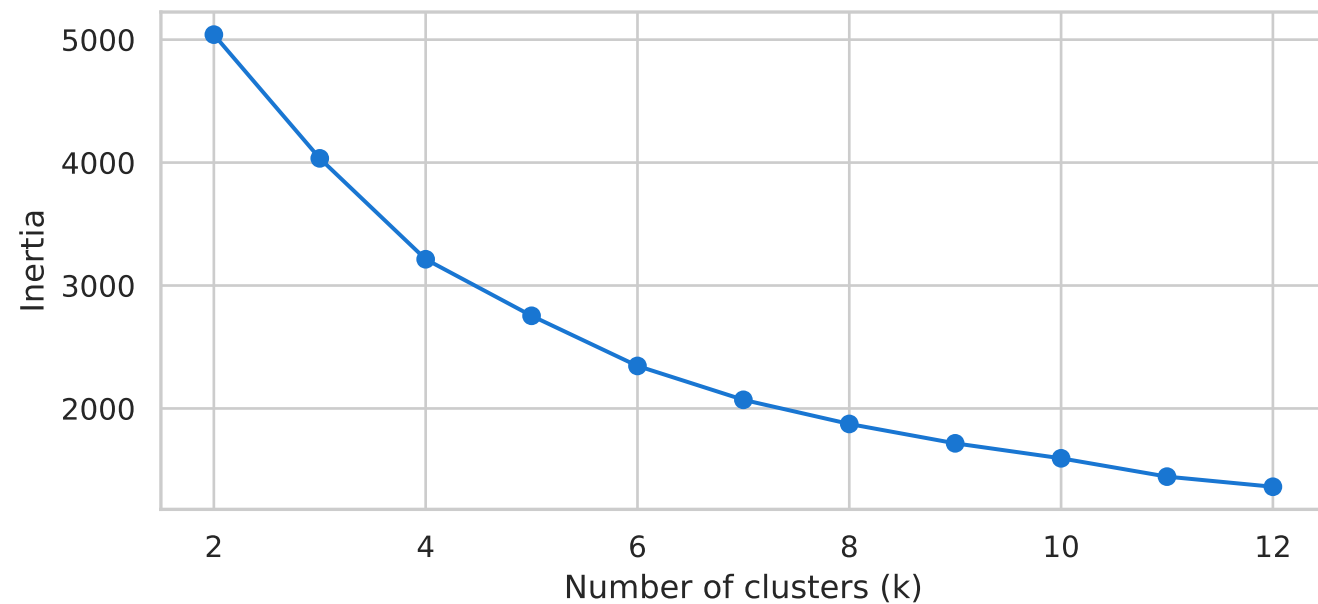
8. K-Means Silhouette Scores by k

Silhouette scores (higher = better separated clusters):

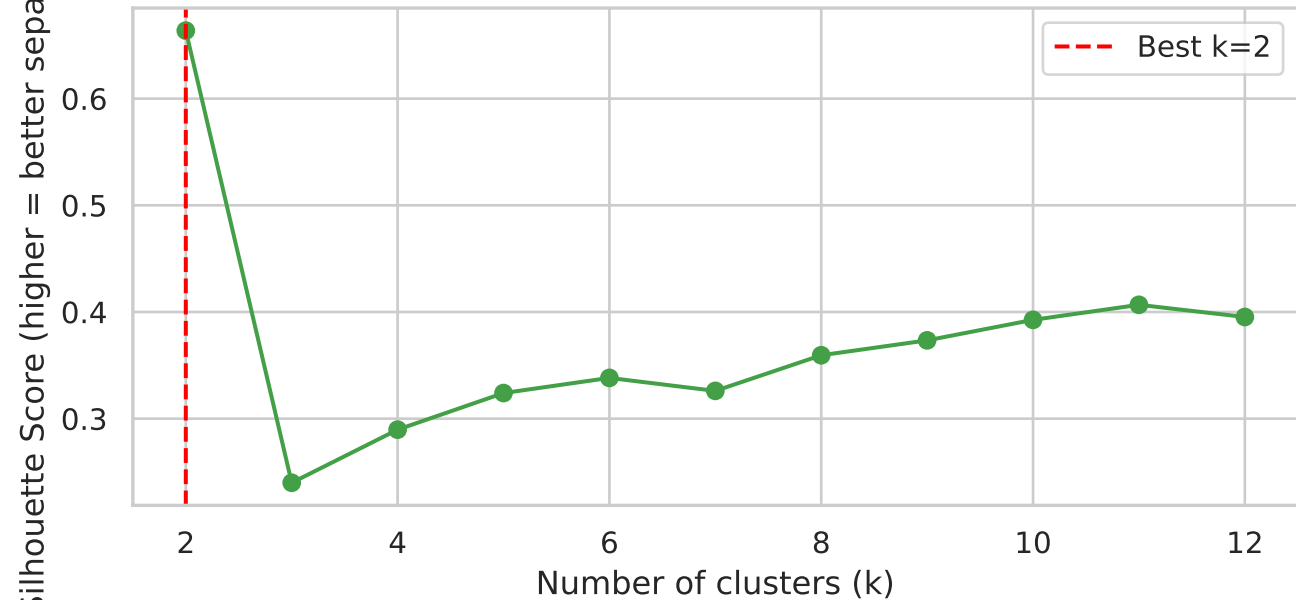
k= 2	0.6637	← best
k= 3	0.2399	
k= 4	0.2897	
k= 5	0.3240	
k= 6	0.3382	
k= 7	0.3261	
k= 8	0.3595	
k= 9	0.3734	
k=10	0.3926	
k=11	0.4067	
k=12	0.3953	

K-Means — Choosing Optimal k

Elbow Curve



Silhouette Score — picks optimal k objectively

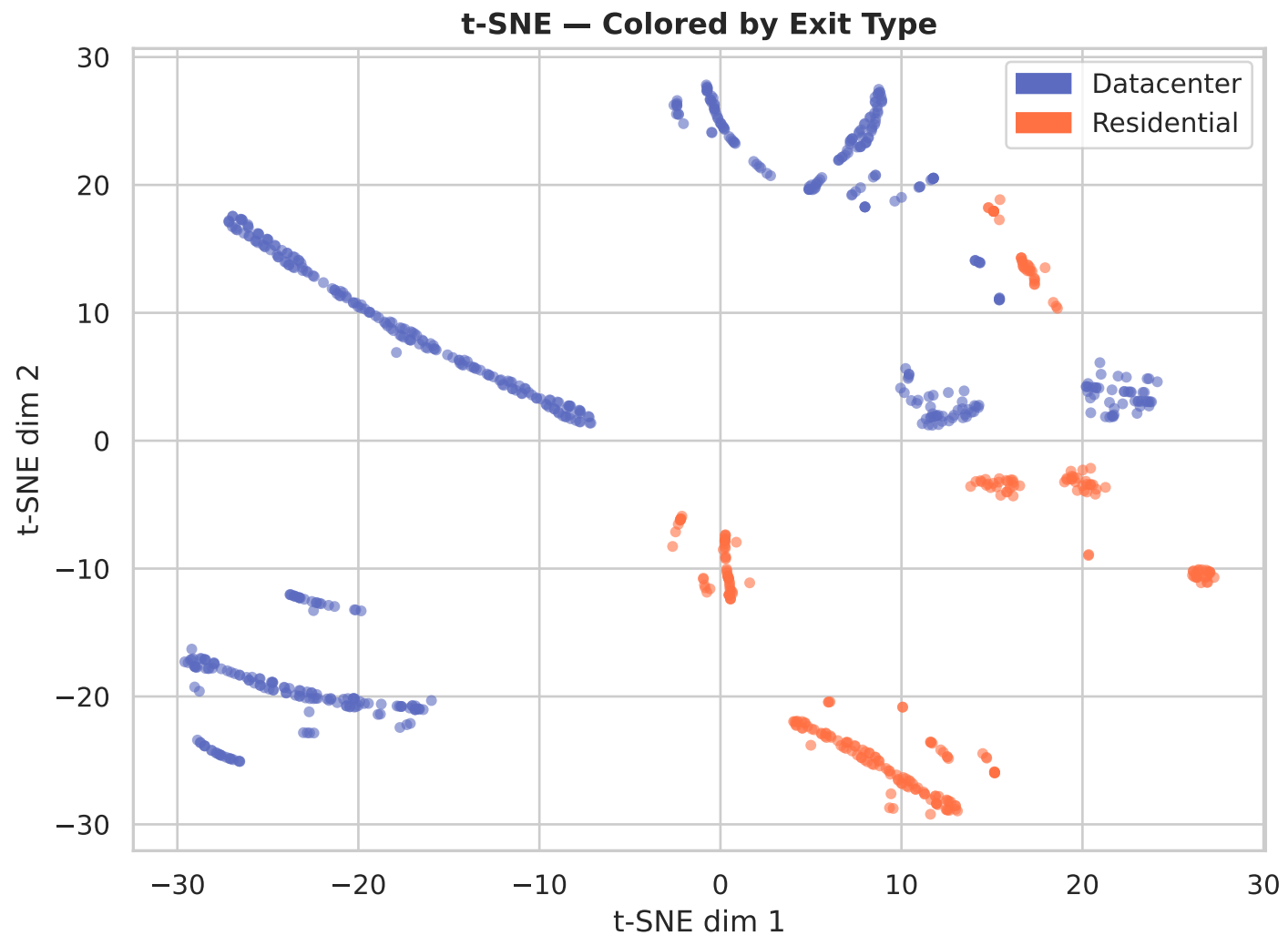
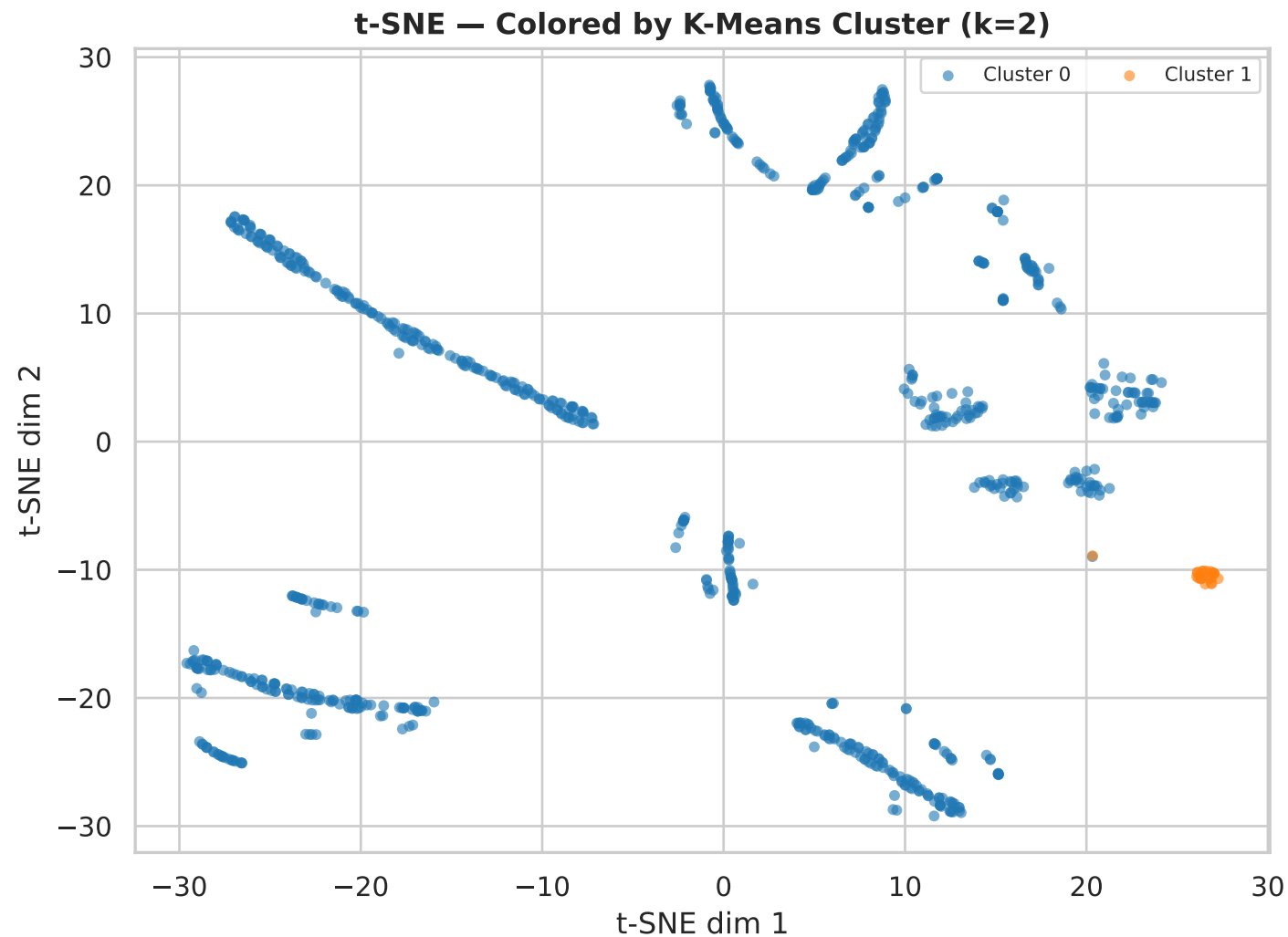


8b. Cluster Profile (k=2)

k=2 selected by silhouette score

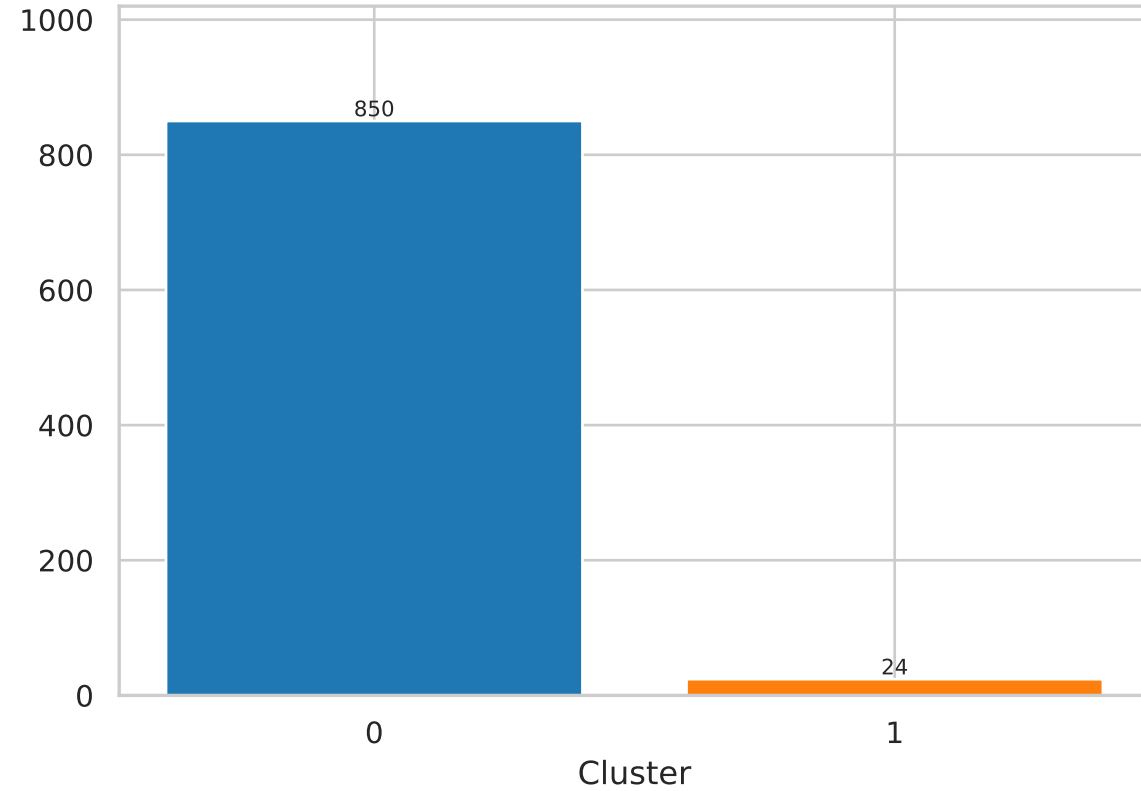
	count	avg_latency	pct_datacenter	pct_blacklist	top_protocol	top_country	pct_proxy_det	pct_ipv6
cluster								
0	850	1951.568	69.6	58.4	vless	Germany	50.2	23.4
1	24	2087.698	0.0	0.0	ss	France	75.0	8.3

Config Space Visualization (t-SNE, working configs only)

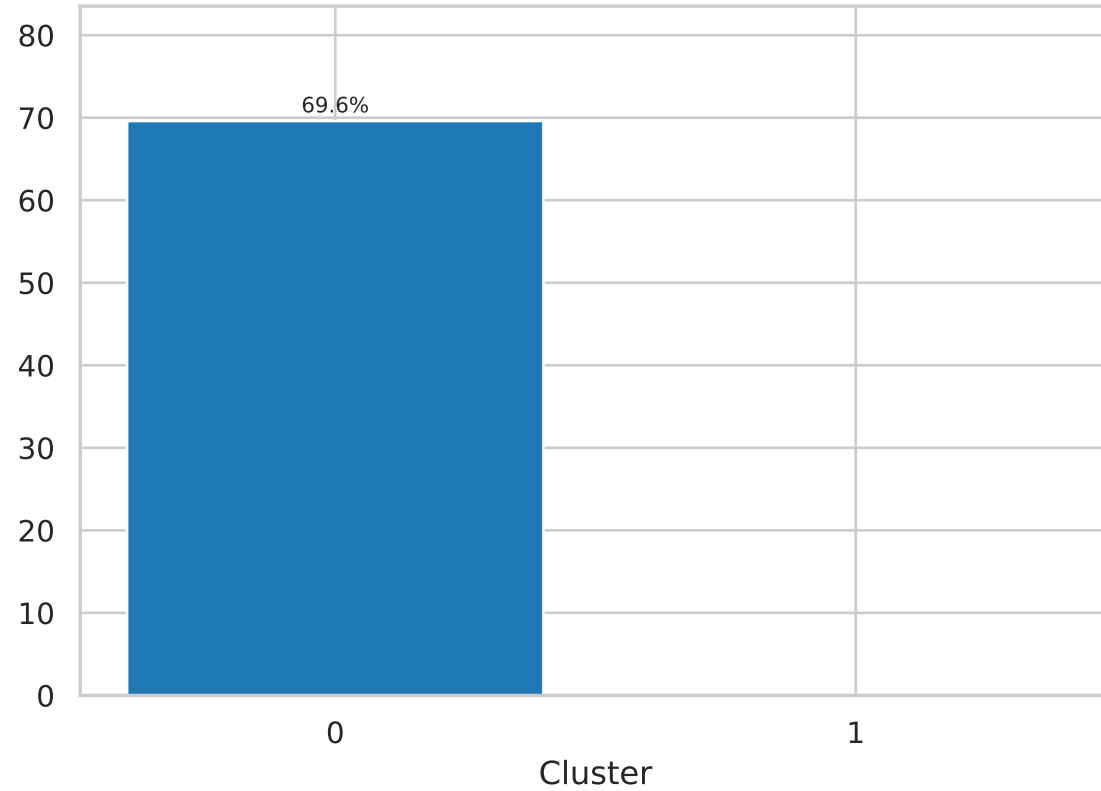


Cluster Profiles — Key Metrics

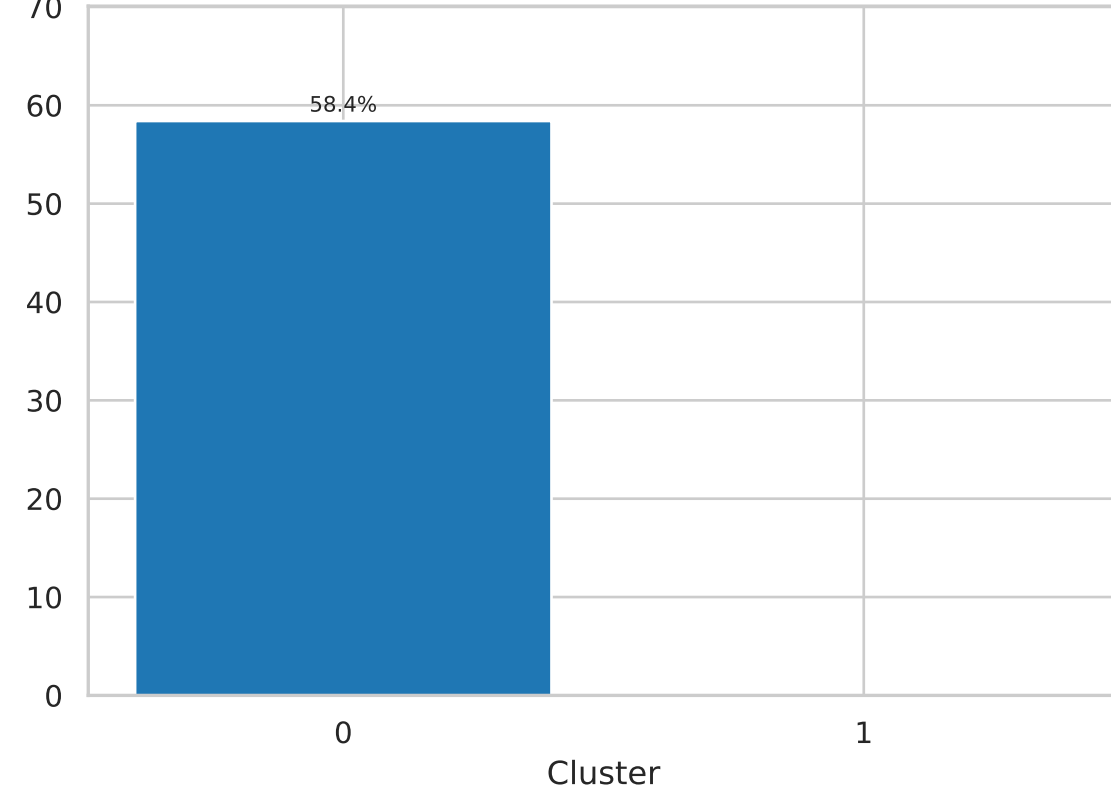
Config Count



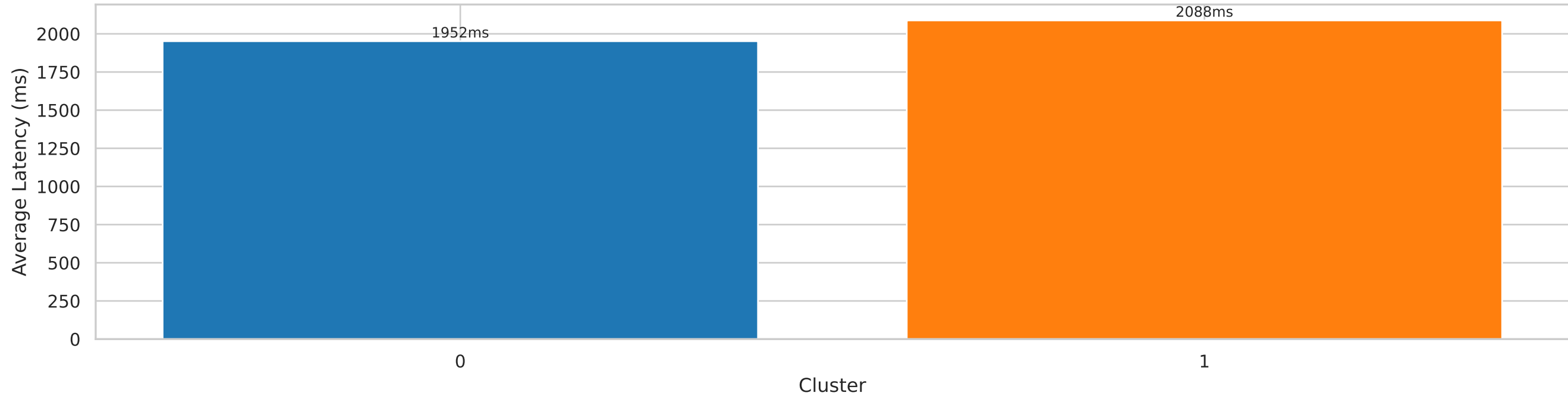
Datacenter Exit (%)



Blacklisted IP (%)



Average Latency per Cluster



11. Anomaly Detection — Breakdown

Normal configs : 830

Anomalous configs: 44

Anomaly breakdown:

Protocol distribution:

protocol

ss 27

hysteria2 7

vless 6

trojan 2

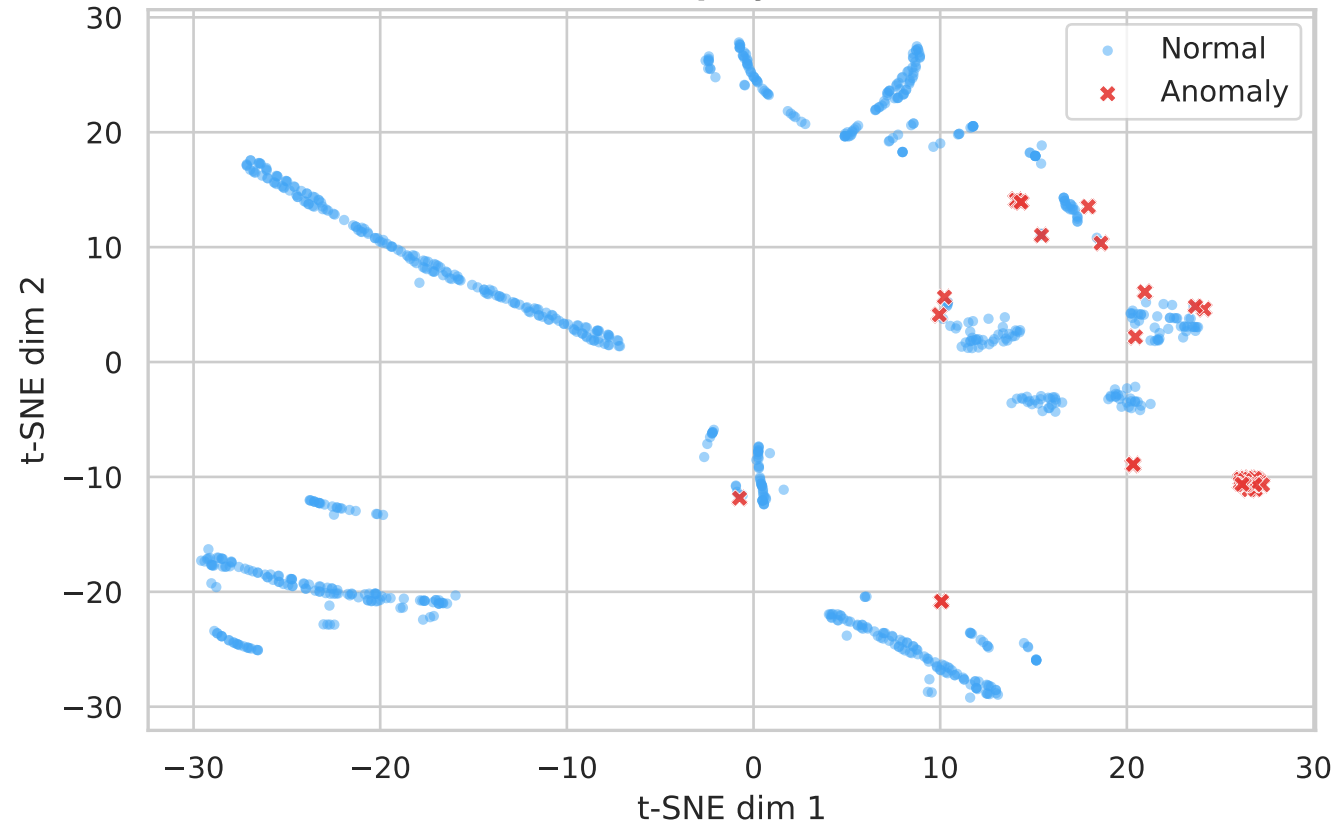
vmess 2

Avg latency : 1803ms (normal: 1963ms)

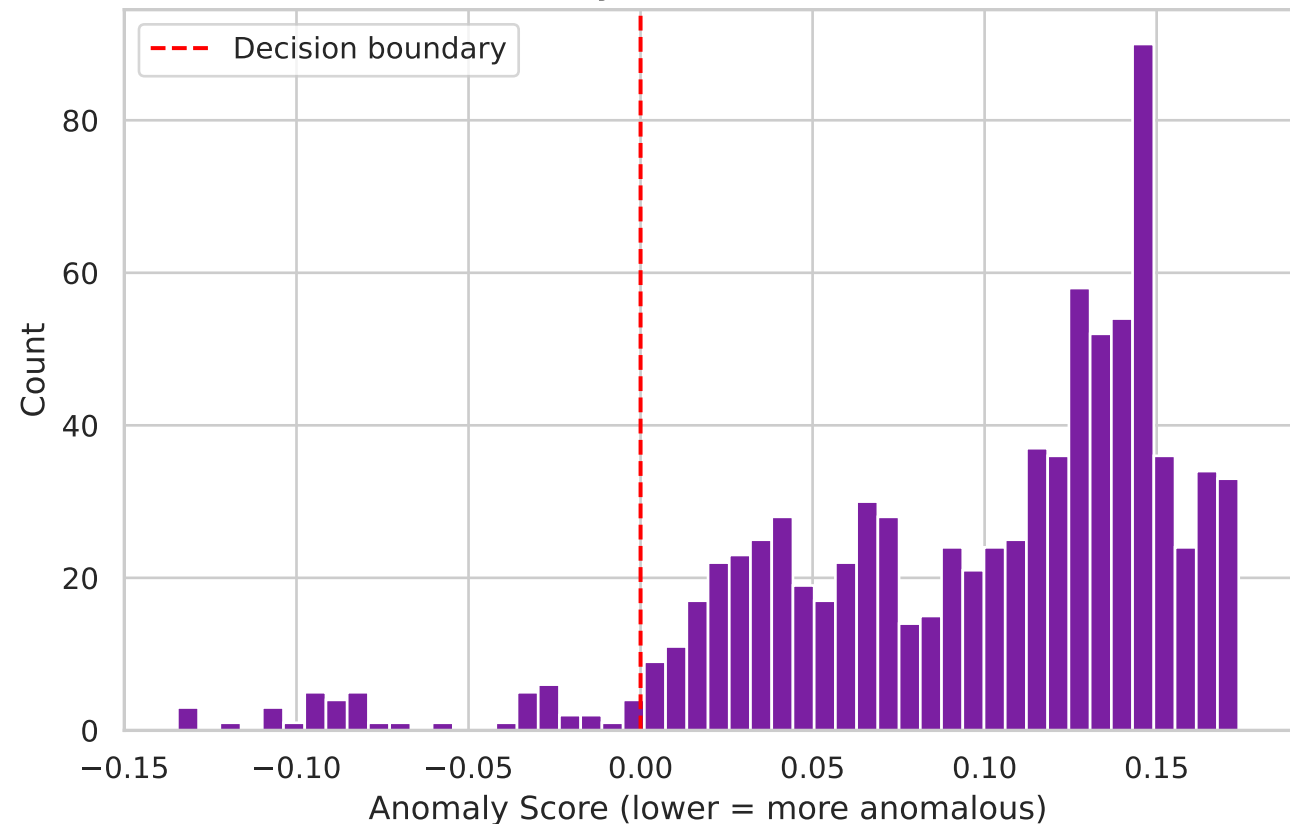
Datacenter pct : 31.8% (normal: 69.6%)

Blacklisted pct: 27.3% (normal: 58.3%)

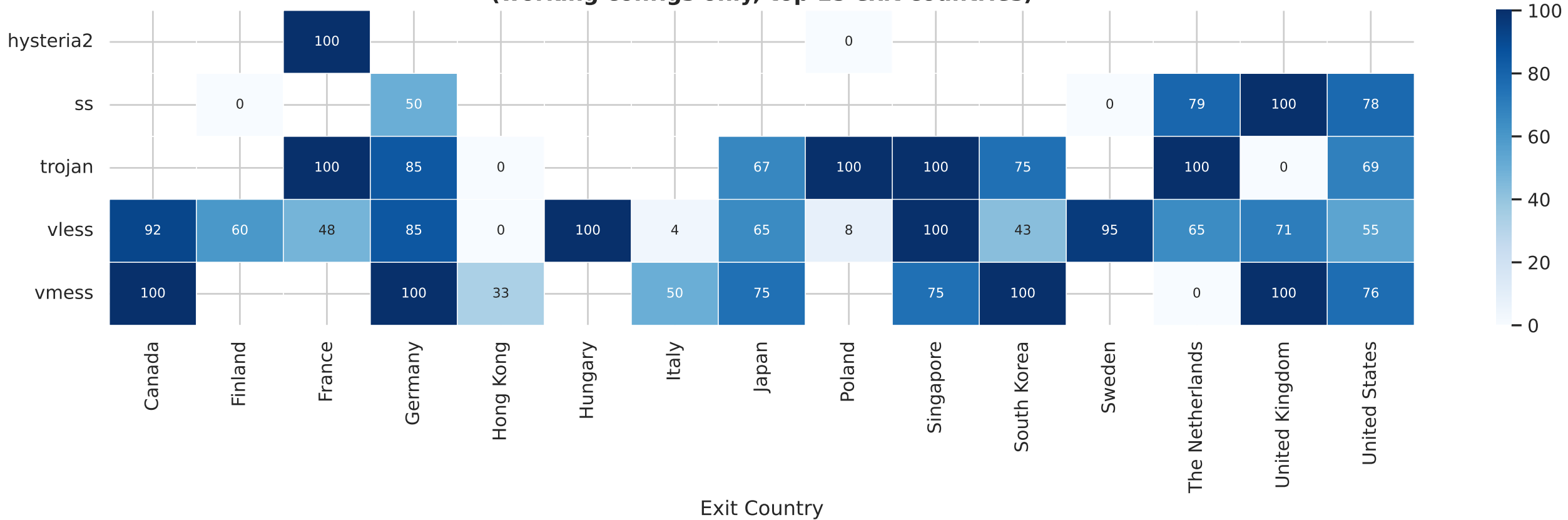
Anomaly Detection (Isolation Forest)
t-SNE projection



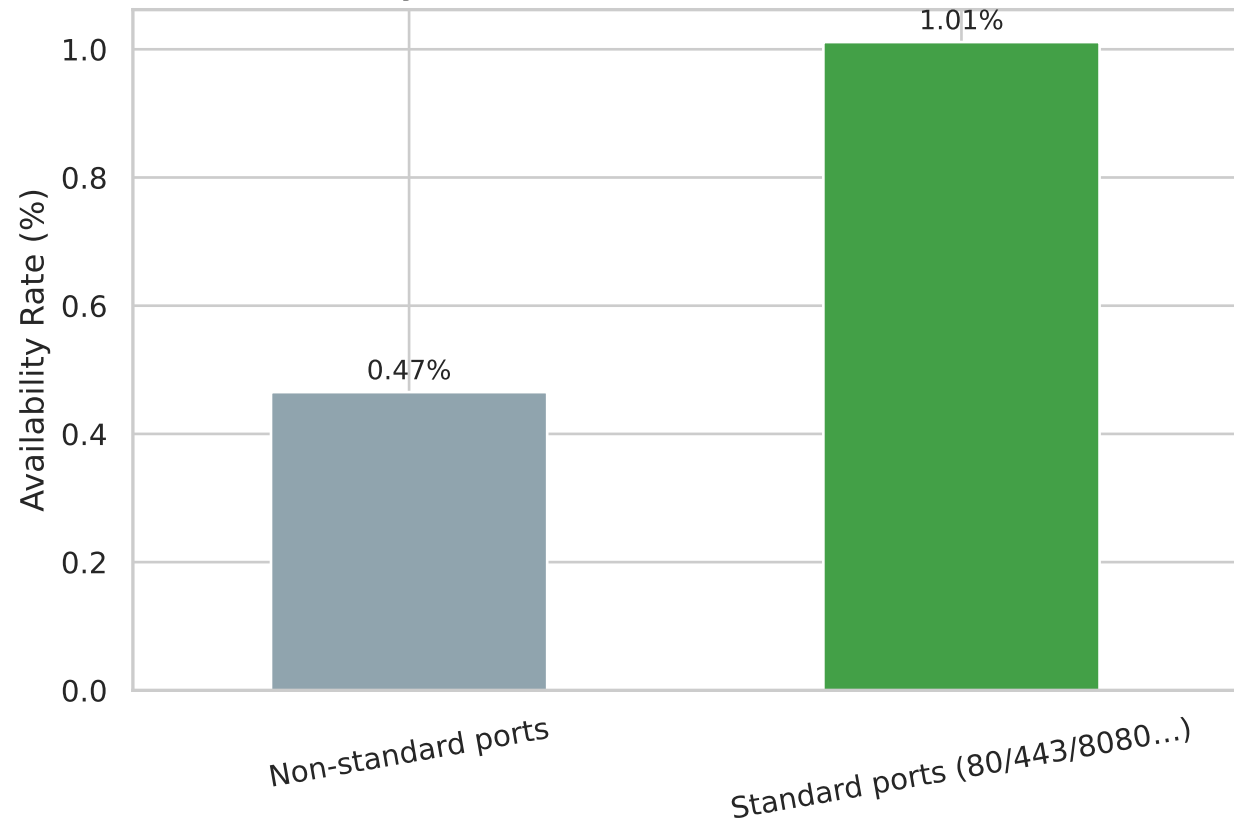
Anomaly Score Distribution



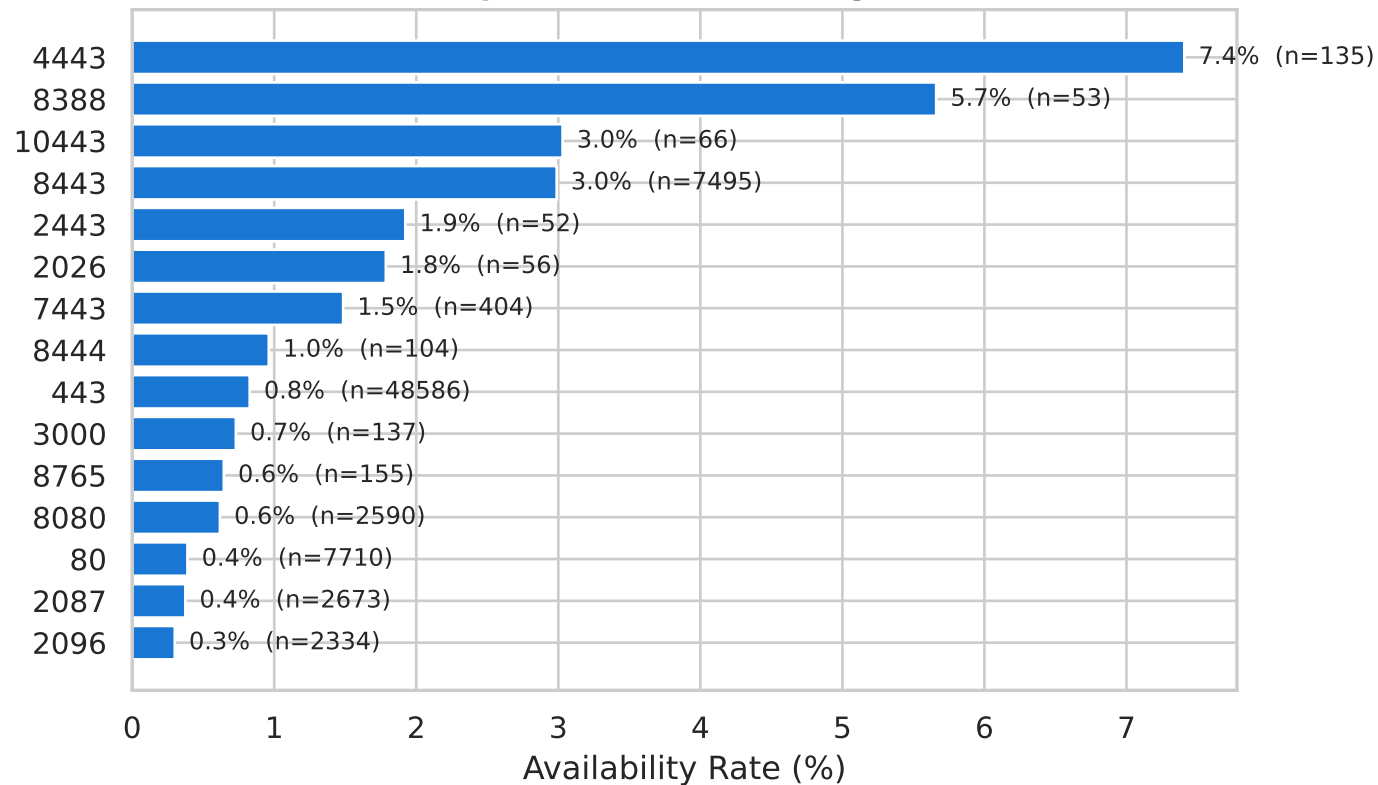
Datacenter Exit Rate (%) — Protocol × Country
(working configs only, top 15 exit countries)



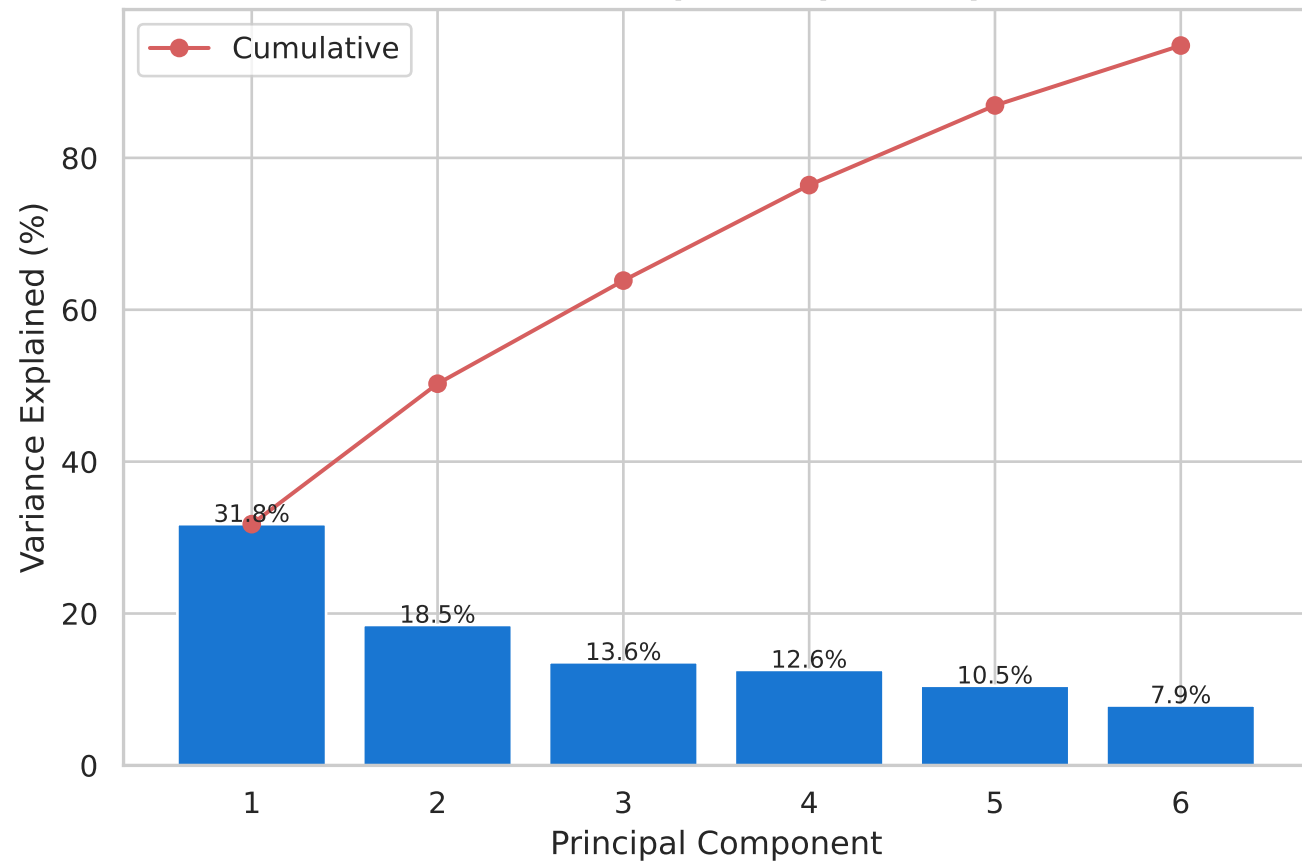
Availability Rate — Standard vs Non-standard Ports



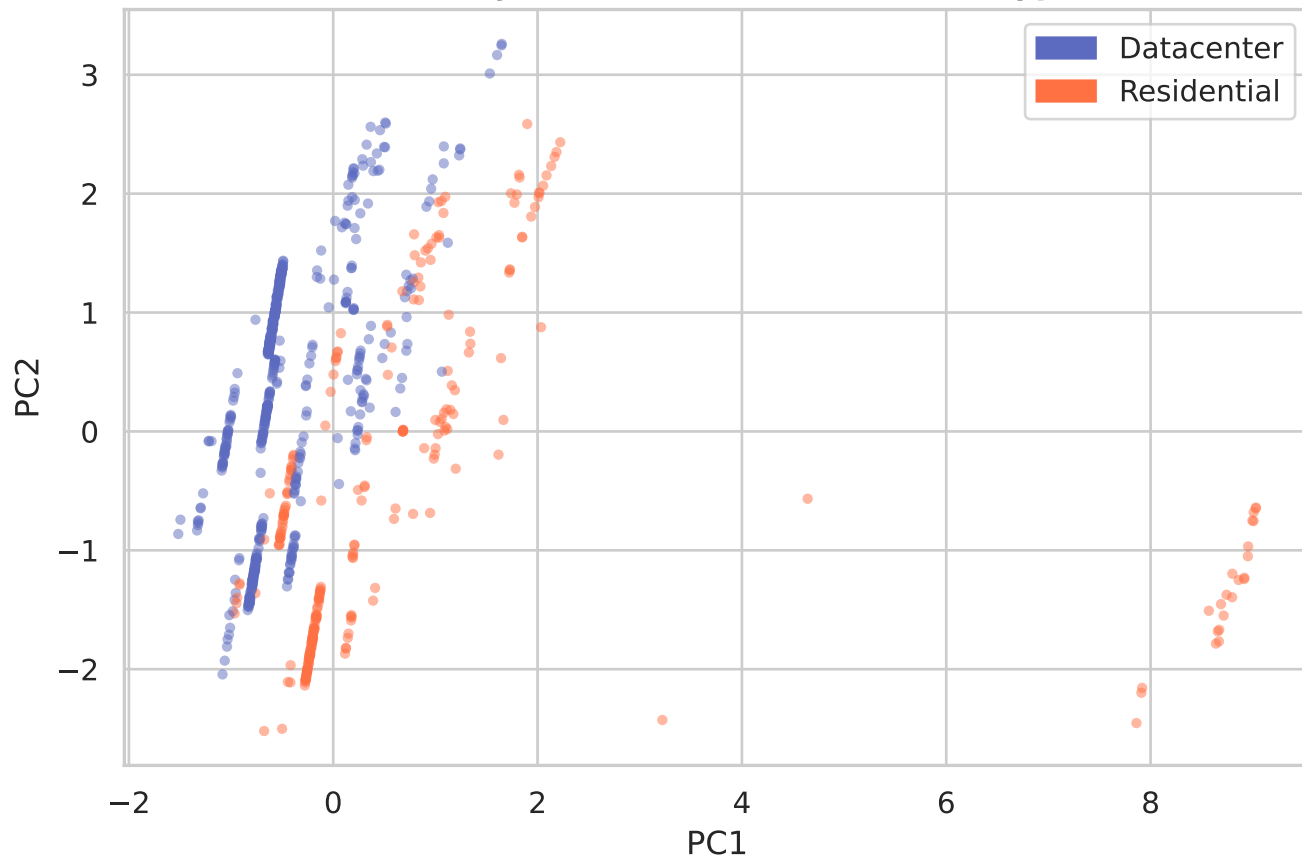
Top 15 Ports by Availability Rate
(ports with ≥50 configs)



PCA — Variance Explained per Component



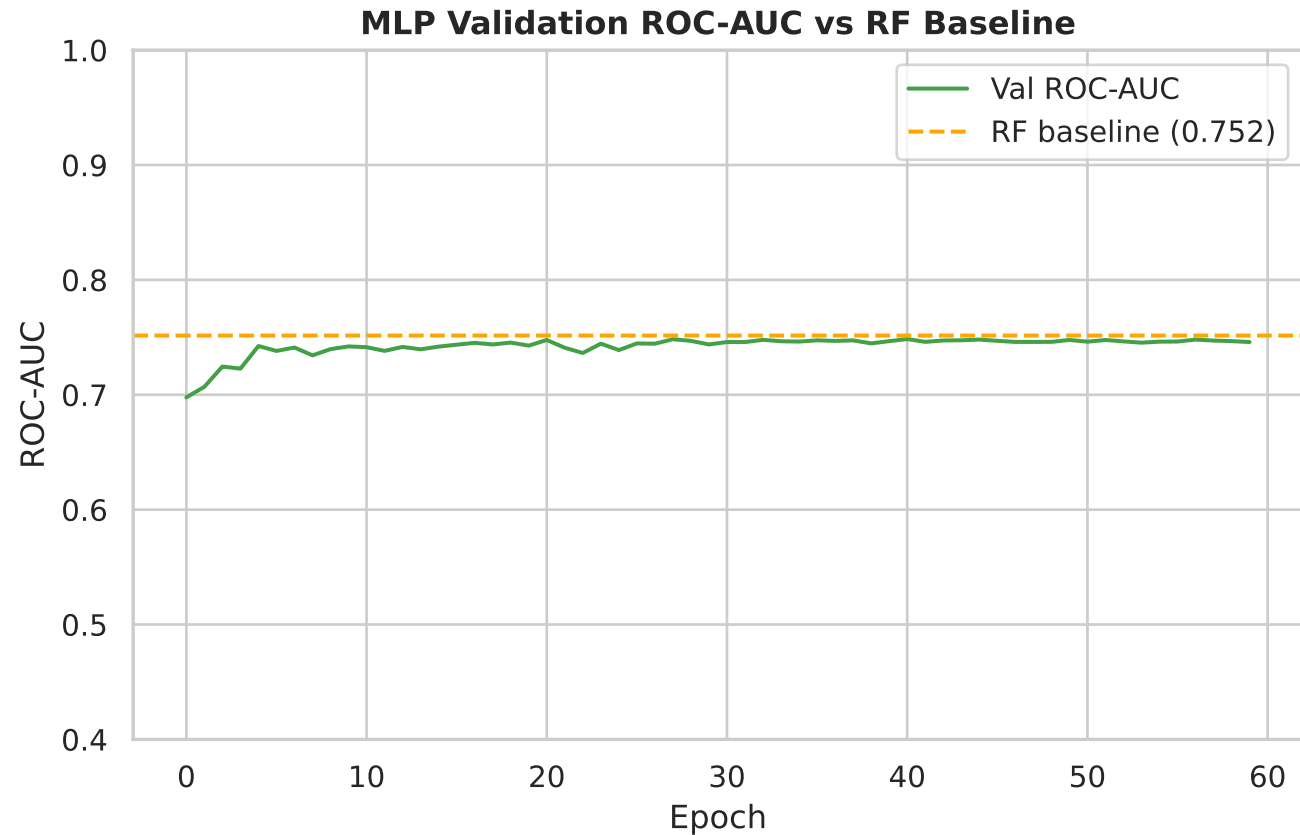
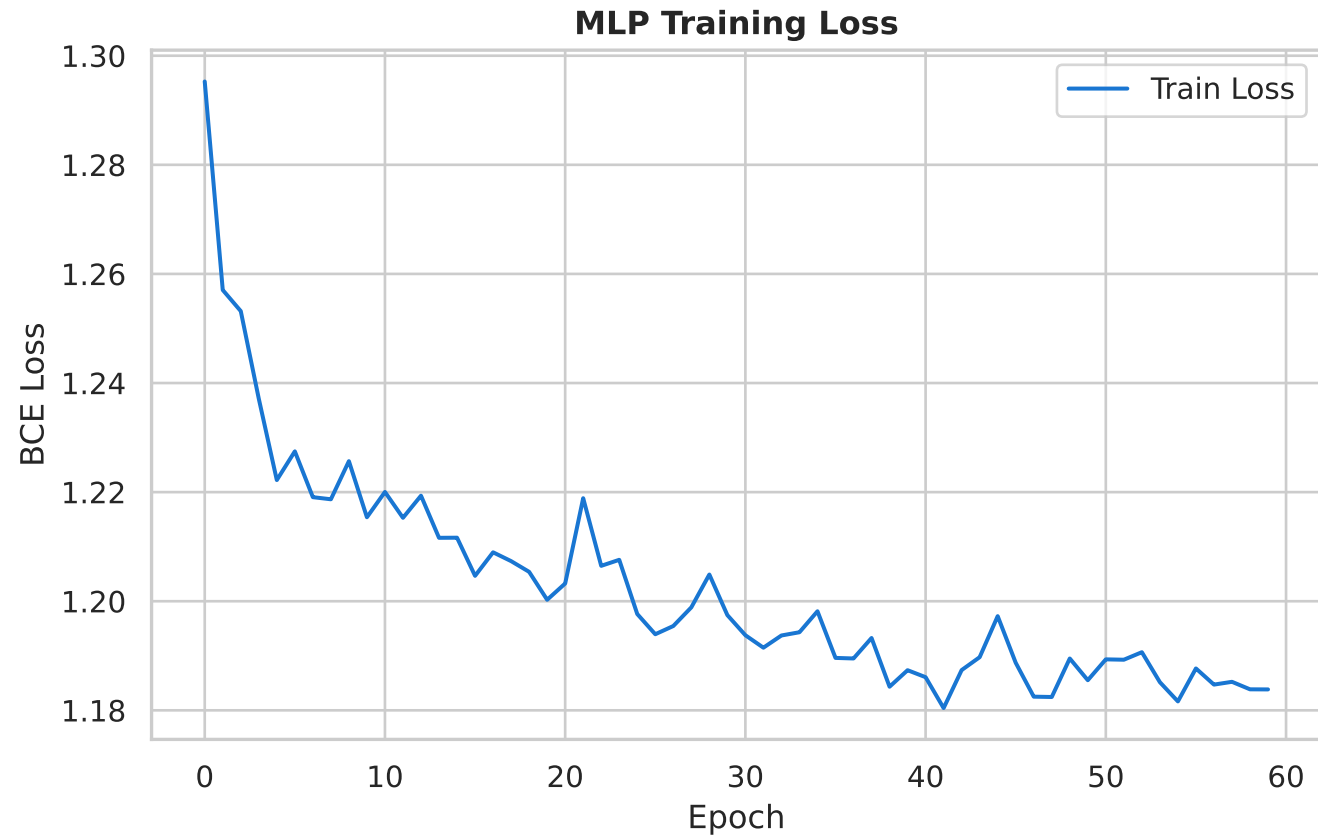
PCA Projection (PC1 vs PC2) — Exit Type



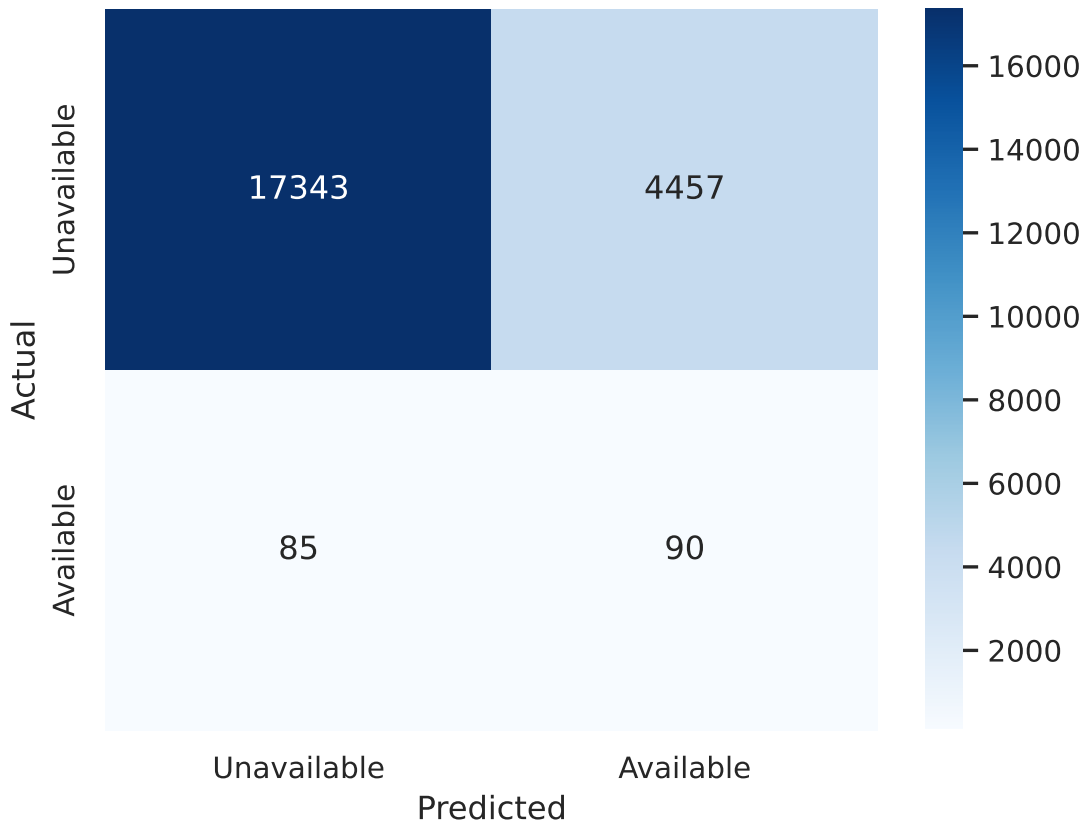
14. PCA Feature Loadings — PC1 and PC2

	PC1	PC2
country_freq	0.563216	-0.151605
asn_freq	0.562065	-0.158722
port_log	0.264479	0.576732
latency_log	-0.041493	-0.219524
is_blacklisted	-0.125978	0.581809
is_datacenter	-0.261364	0.297390
protocol_enc	-0.307813	-0.143427
is_standard_port	-0.340984	-0.351424

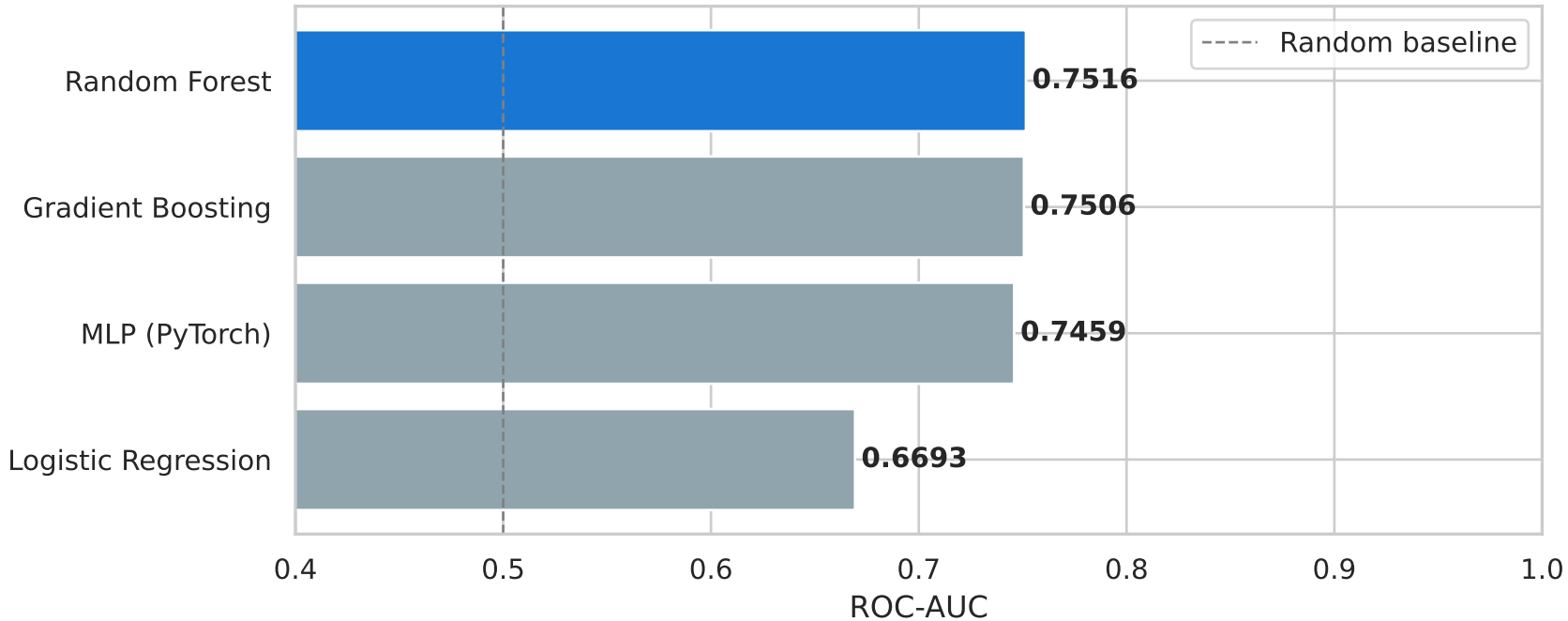
MLP Availability Classifier — Training Curves



Confusion Matrix — MLP Availability Classifier
(Test ROC-AUC = 0.746)



Availability Prediction — Model Comparison (ROC-AUC)



17. Final Model Ranking — Availability Prediction

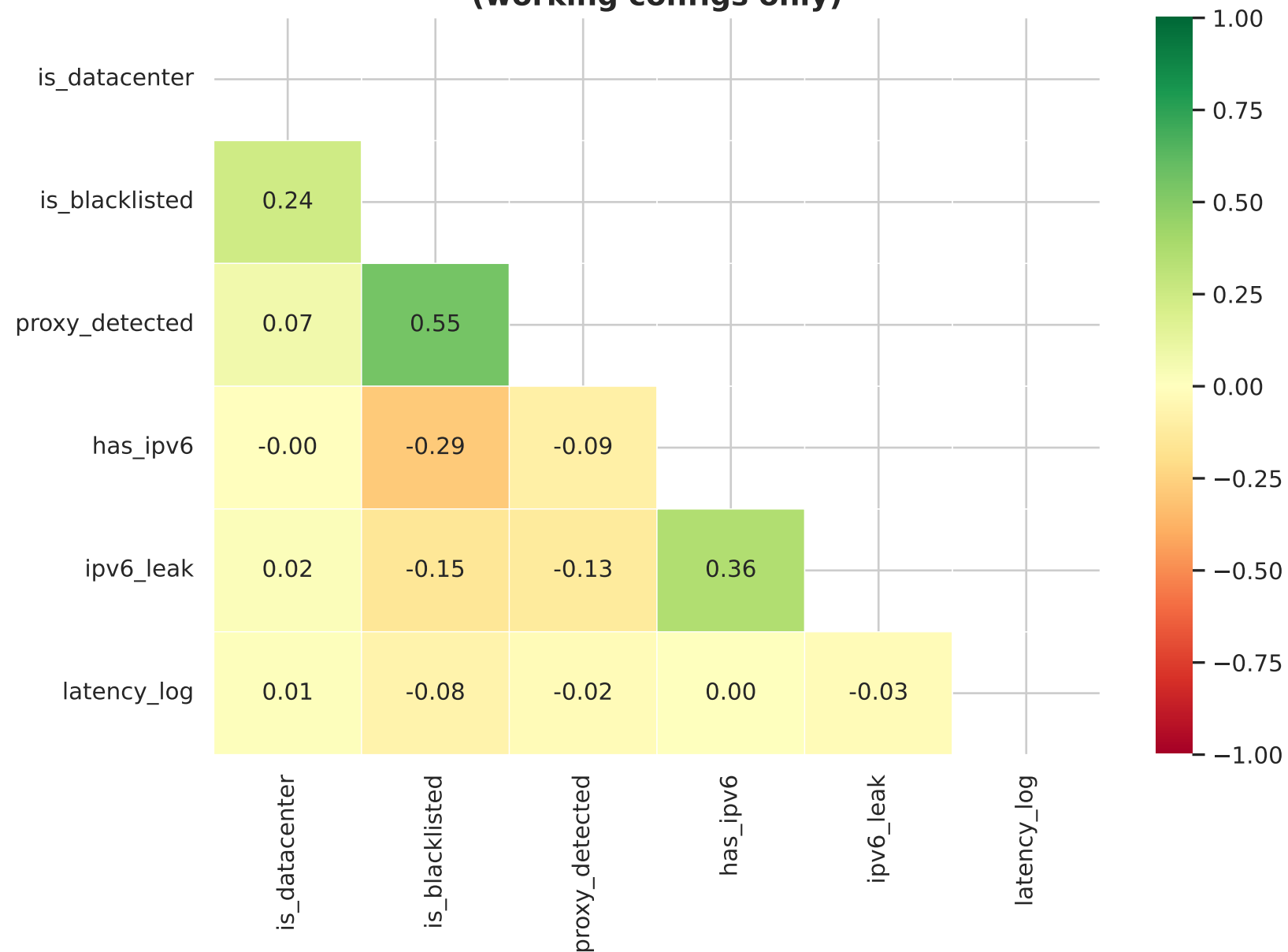
Final Model Ranking — Availability Prediction (by ROC-AUC):

1. Random Forest	0.7516
2. Gradient Boosting	0.7506
3. MLP (PyTorch)	0.7459
4. Logistic Regression	0.6693

Note: ROC-AUC > 0.5 = better than random guessing.

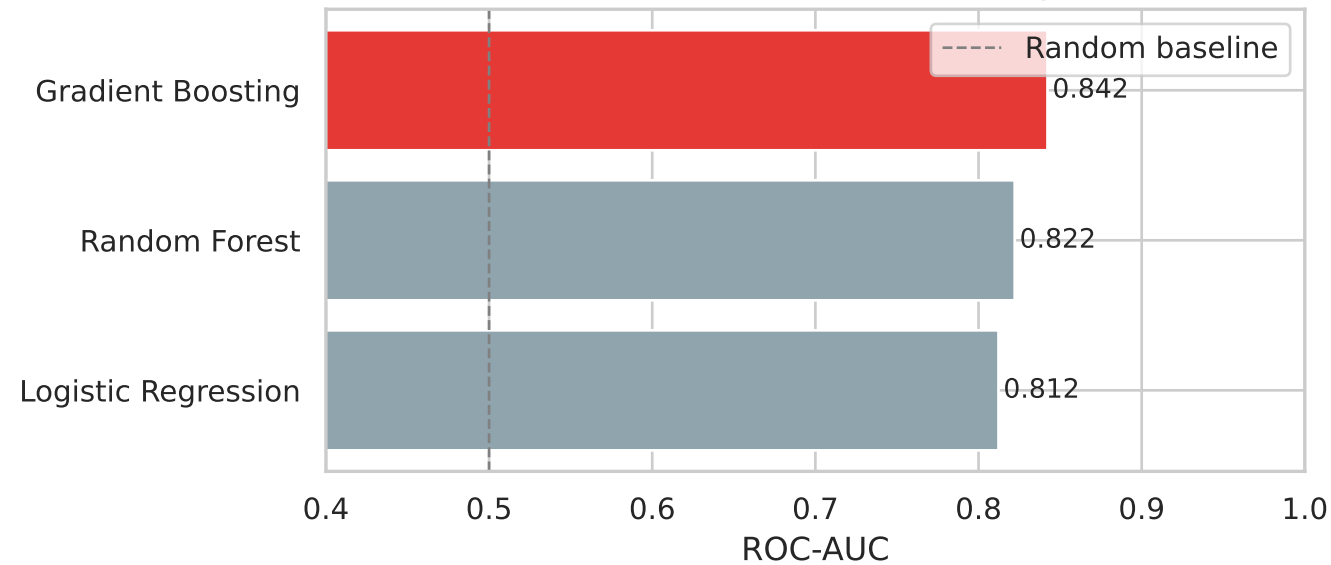
Static config metadata (protocol + port) has limited but non-trivial predictive power for availability.

Security Feature Correlations (working configs only)

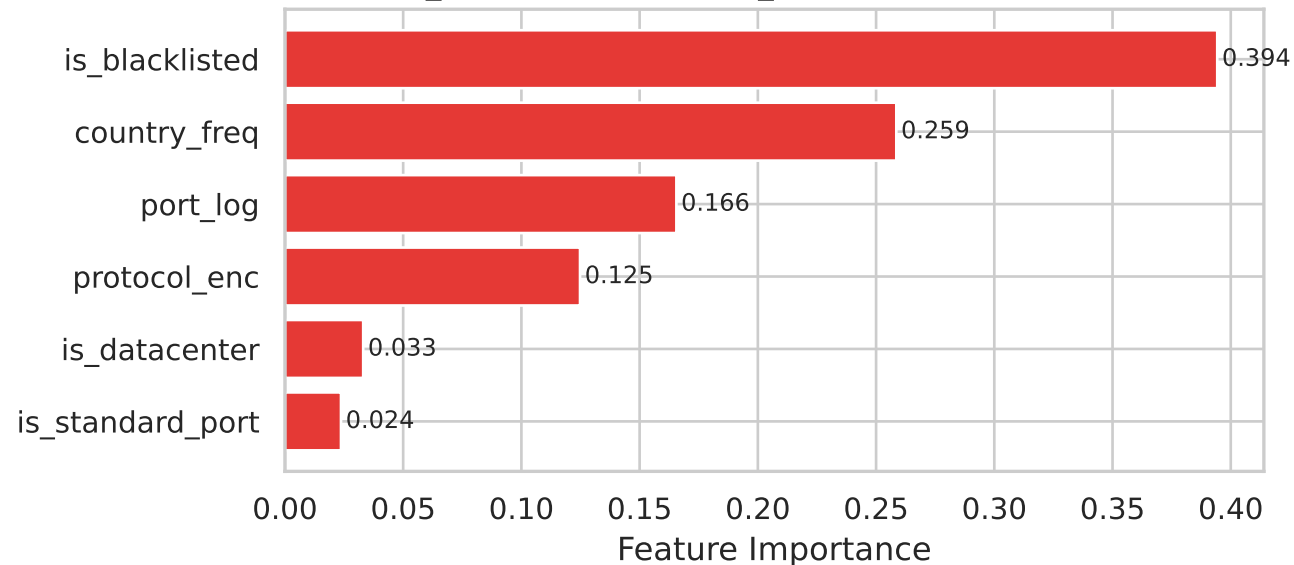


Predicting Whether a Config Will Be Flagged as a Proxy

**Predicting Proxy Detection
(can we know before testing?)**

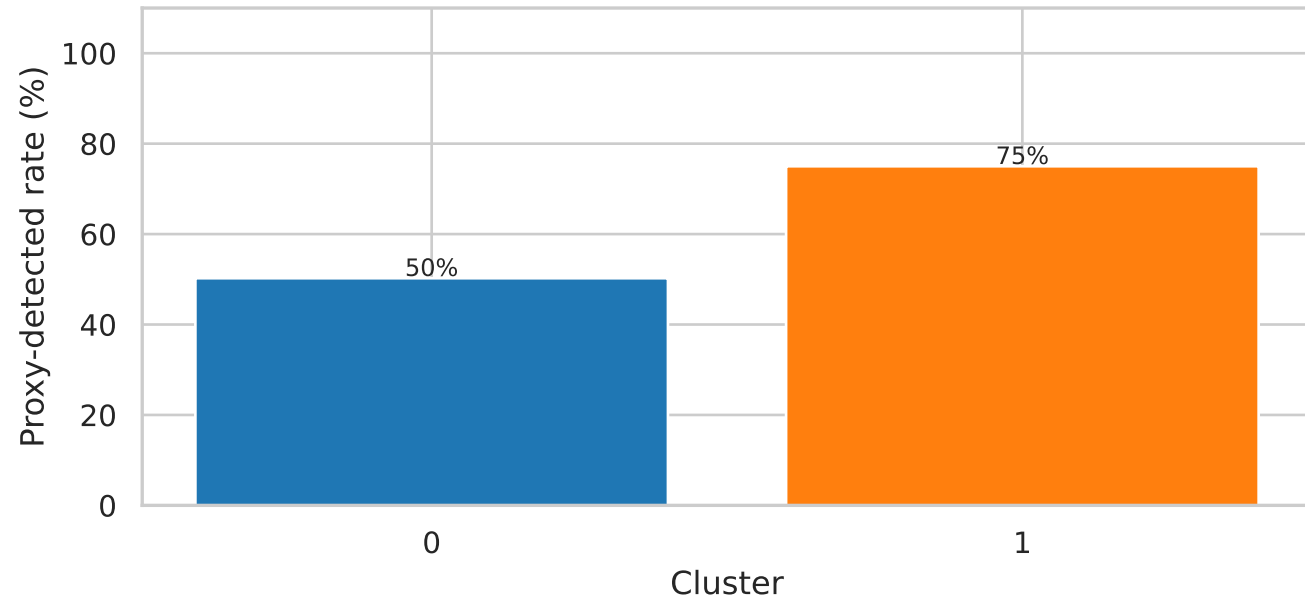


**Proxy Detection — Feature Importance
(is_blacklisted and is_datacenter dominate)**



Security Profile by K-Means Cluster

Proxy Detection Rate per Cluster



IPv6 Support Rate per Cluster

